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Lab: Lab 6

Date: 4/24/2022

a. Program Description:

The purpose of this is to create a dictionary. This dictionary is then passed on to a Json file, which is first created and contents of dictionary are loaded into it. Once the Json file is created, a dictionary is then used to read its contents. The values of the dictionary are then searched to find required values within it.

In this program, we have learnt the concept of dictionary, nested dictionary and reading their values, and json files.

b. Program Source Code:

```
import json
import print_me_first

def create_my_contact():
    my_contact = {"01":
        {
            "firstName": "John", "lastName": "Smith", "DOB": "1/20/1991",
            "phoneNum": {"number": "510-600-5400", "type": "cell"},
            "address": {"street": "100 main street", "city": "Fremont", "state": "CA", "zipcode": "94536"}
        },
        "02":
        {
            "firstName": "Ron", "lastName": "Robertson", "DOB": "5/23/1991",
            "phoneNum": {"number": "510-600-8800", "type": "cell"},
            "address": {"street": "4600 Ohlone Way", "city": "Fremont", "state": "CA", "zipcode": "94539"}
        },
        "03":
        {
            "firstName": "Paul", "lastName": "Washington", "DOB": "6/15/1995",
            "phoneNum": {"number": "510-688-1241", "type": "cell"},
            "address": {"street": "8543 Ohlone Plaza", "city": "Fremont", "state": "CA", "zipcode": "94539"}
        }
    }
    return(my_contact)

def save_json_file(filename, json_object):
    with open(filename, "w") as outfile:
        outfile.write(json.dumps(json_object))
    outfile.close()

def open_json_file(filename):
    with open(filename, "r") as jsonfile:
        datafile = json.load(jsonfile)
    return datafile
```

```

def find_my_contact_key(key, json_data):
    print("*** Searching for ", key)
    flag = False
    for items in json_data:
        for entry in json_data[items]:
            if key == json_data[items][entry]:
                print("*** %s" %key, "found" " ***")
                name = json_data[items]["firstName"] + " " + json_data[items]["lastName"]
                birthday = json_data[items]["DOB"]
                cell = json_data[items]["phoneNum"]["number"]
                street = json_data[items]["address"]["street"]
                city = json_data[items]["address"]["city"]
                state = json_data[items]["address"]["state"]
                zipcode = json_data[items]["address"]["zipcode"]
                print("Name: %20s" % (name))
                print("Birthday: %12s" % (birthday))
                print("cell: %19s" % (cell))
                print("Address: %19s \n %19s, %s %s \n" % (street, city, state, zipcode))
                flag = True
            else:
                continue
    if not flag:
        any = " "
        print("*** %s" %key, "not found" " ***")

```

```

print("*** %s" %key, "not found" " ***")

if __name__ == "__main__":
    print_me_first.print_me_time()
    contact = create_my_contact()
    jsonwrite = json.dumps(contact, indent=2)
    save_json_file("my_contact.json", jsonwrite)
    json_data = open_json_file("my_contact.json")
    print("***BEGINNING OF JSON List: \n", json_data, "\n*** END OF JSON LIST\n\n")
    find_my_contact_key("Ron", json_data)
    find_my_contact_key("Sha", json_data)

```

```

create_my_contact.py x my_contact.json x dictionary_json_example.py x decoder.py x print_me_first.py x printuser.py x Mortgage.py x InterestCalculator.py x
1 import os.path
2 import sys
3 from datetime import datetime
4
5 def print_me_time():
6     #This program is to print name, lab name and current time and the message Hello World
7     name = "Shweta Chauhan - CNET 142 - Lab 6 Dictionary"
8     #Printing Name based on the variable defined as name
9     print("{:14}".format("Name"), ":", name)
10    program = os.path.basename(sys.argv[0])
11    #Printing Lab name based on the variable defined as lab
12    print("{:14}".format("Program"), ":", program)
13    #Taking current date and time
14    currentTime = datetime.now()
15    #print(currentTime)
16    timestamp_str = currentTime.strftime("%b-%d-%Y %a (%I:%M:%S%p)")
17    #Printing current date and time based on the variables defined as timestamp_str
18    print("{:14}".format("Current Time"), ":", timestamp_str)
19
20

```

c. Program Output

```
create_my_contact
C:\Users\sheetachauhan\AppData\Local\Programs\Python\Python310\python.exe C:/Users/sheetachauhan/PycharmProjects/Course_scripts/create_my_contact.py
Name      : Sheetachauhan - CNEF 142 - Lab 8 Dictionary
Program   : create_my_contact.py
Current Time : Apr-24-2022 Sun (11:00:51PM)

***BEGINNING OF JSON List:
{'01': {'firstName': 'John', 'lastName': 'Smith', 'DOB': '1/20/1991', 'phoneNum': {'number': '510-600-5400', 'type': 'cell'}, 'address': {'street': '100 main street', 'city': 'Fremont', 'state': 'CA', 'zipcode': '94536'}},
{'02': {'firstName': 'Ron', 'lastName': 'Robertson', 'DOB': '5/23/1991', 'phoneNum': {'number': '510-600-8800', 'type': 'cell'}, 'address': {'street': '4600 Ohlone Way', 'city': 'Fremont', 'state': 'CA', 'zipcode': '94539'}},
{'03': {'firstName': 'Paul', 'lastName': 'Washington', 'DOB': '6/15/1995', 'phoneNum': {'number': '510-688-1241', 'type': 'cell'}, 'address': {'street': '8543 Ohlone Plaza', 'city': 'Fremont', 'state': 'CA', 'zipcode': '94539'}}}

*** END OF JSON LIST

*** Searching for Ron
*** Ron found ***
Name:      Ron Robertson
Birthday:  5/23/1991
cell:      510-600-8800
Address:   4600 Ohlone Way
           Fremont, CA 94539

*** Searching for Sha
*** Sha not found ***
```

Json file Output:

```
{
  "01": {
    "firstName": "John",
    "lastName": "Smith",
    "DOB": "1/20/1991",
    "phoneNum": {
      "number": "510-600-5400",
      "type": "cell"
    },
    "address": {
      "street": "100 main street",
      "city": "Fremont",
      "state": "CA",
      "zipcode": "94536"
    }
  },
  "02": {
    "firstName": "Ron",
    "lastName": "Robertson",
    "DOB": "5/23/1991",
    "phoneNum": {
      "number": "510-600-8800",
      "type": "cell"
    },
    "address": {
      "street": "4600 Ohlone Way",
      "city": "Fremont",
      "state": "CA",
      "zipcode": "94539"
    }
  },
  "03": {
    "firstName": "Paul",
    "lastName": "Washington",
    "DOB": "6/15/1995",
    "phoneNum": {
      "number": "510-688-1241",
      "type": "cell"
    },
    "address": {
      "street": "8543 Ohlone Plaza",
      "city": "Fremont",
      "state": "CA",
      "zipcode": "94539"
    }
  }
}
```

```
},  
"03": {  
  "firstName": "Paul",  
  "lastName": "Washington",  
  "DOB": "6/15/1995",  
  "phoneNum": {  
    "number": "510-688-1241",  
    "type": "cell"  
  },  
  "address": {  
    "street": "8543 Ohlone Plaza",  
    "city": "Fremont",  
    "state": "CA",  
    "zipcode": "94539"  
  }  
}  
}
```